AI CLUSTERS

Network Configurations for Optimal AI Fabrics including MRC and SRv6

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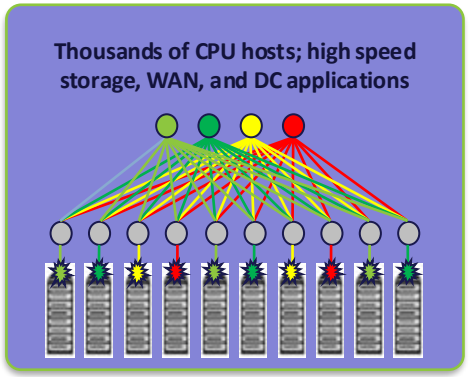
OCP PLATINUM Member

Agenda

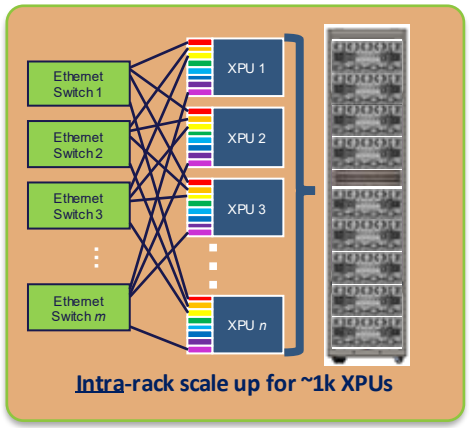
- AI Networks for Training and Inference
- AI Fabrics: Design Choices
- Building Multi-Plane AI Fabric
- Modern Transport for AI Fabric: MRC
- Packet Multipathing
- Source Routing with SRv6
- Active Path Probing
- Advanced Telemetry
- Call to Action
- Q&A

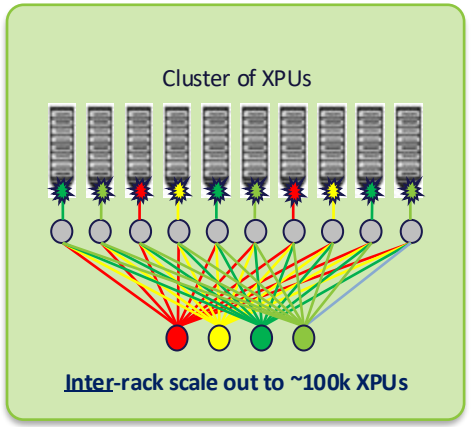
AI Networks for Training and Inference

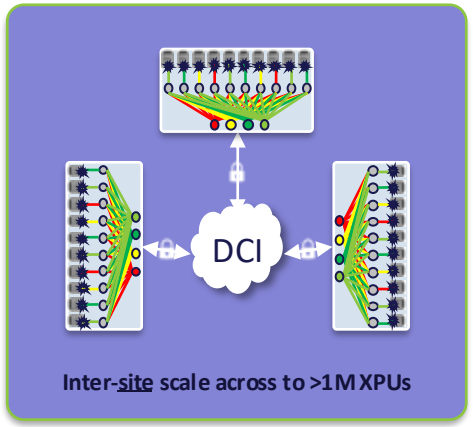
Front End Data Center



The 3 AI Fabrics



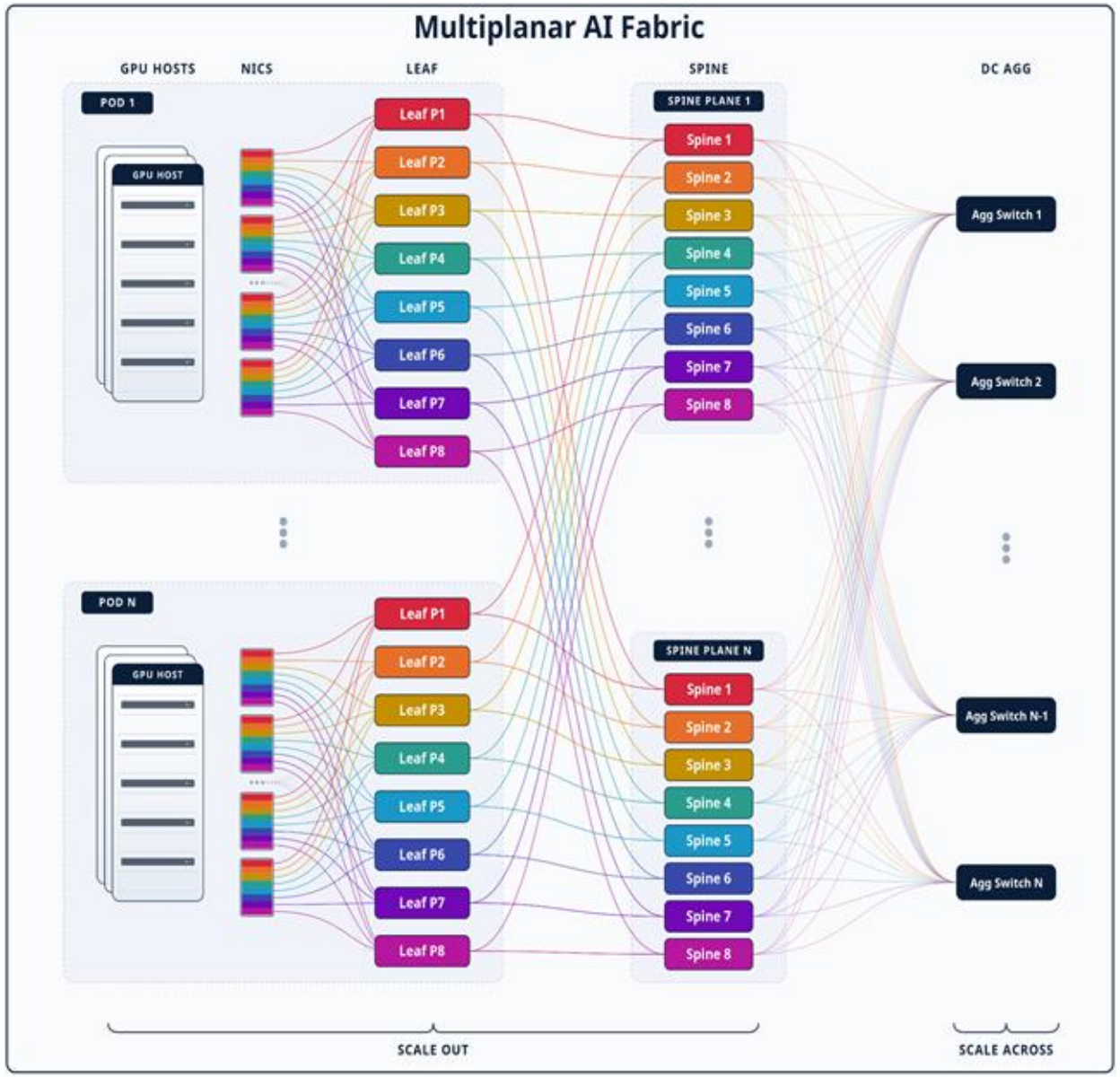




Scale Up | **Scale Out** | **Scale Across**

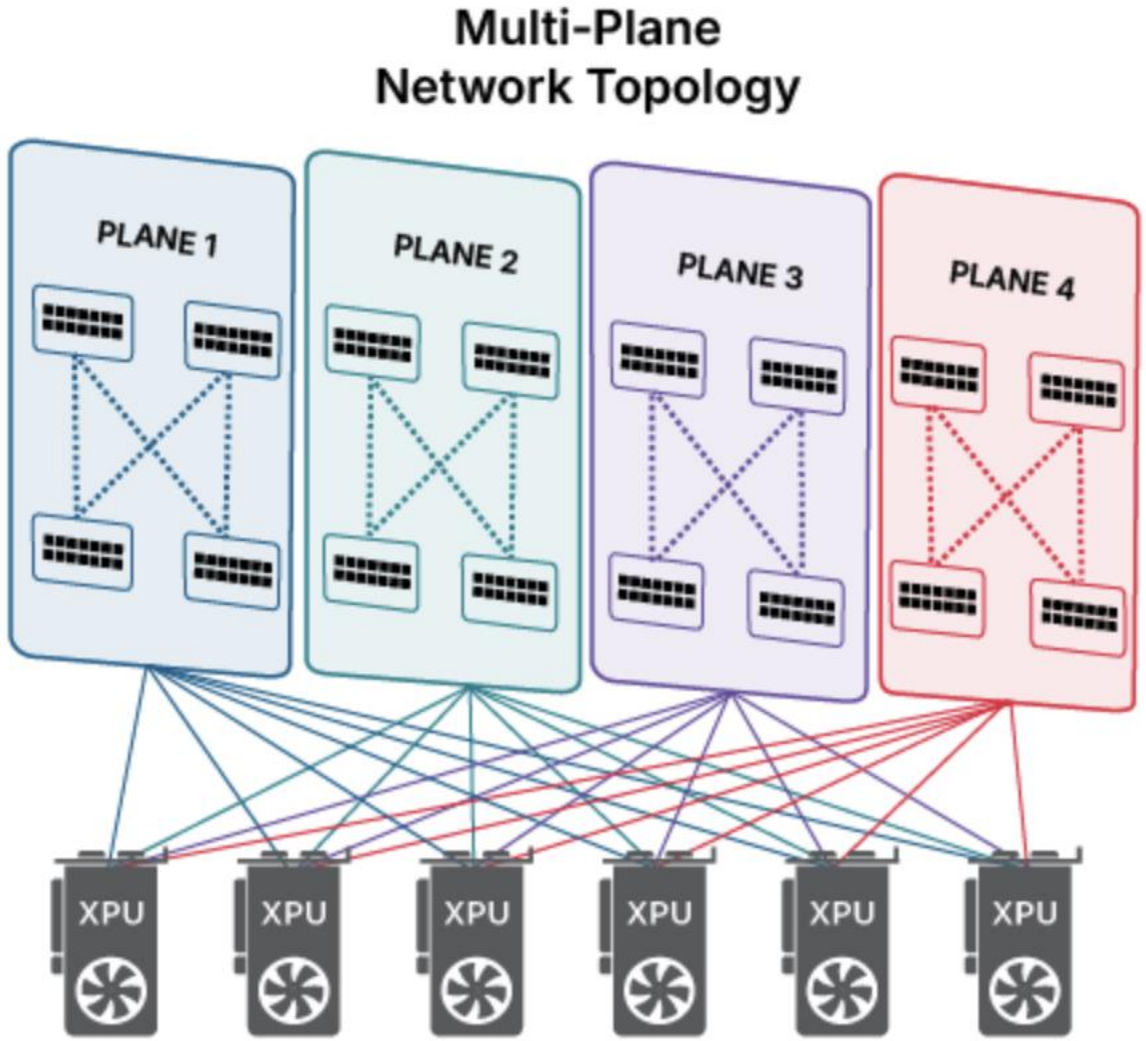
AI Fabrics: Design Choices

- Technology
  - Endpoint Scheduled/Non-Scheduled Fabric (NSF)
    - Low Latency , higher switch capacity
    - Simplified Cabling
  - Switch Scheduled Fabric / Disaggregated Switch Fabric (DSF)
    - NIC Agnostic
    - Deep Buffer
  - Mix of shallow buffer and deep buffers
- Topology
  - Similar topology choices possible with both Technology Options :
  - Top of Rack (ToR)/End of Row(EoR)/Middle of Row (MoR)
  - Rail Based/Non-Rail, Single/Multi-Planar
  - Single/Two/Three Stage Clos Fabric



Multi-Plane AI Fabrics

- Fault resilient
- Better Load Balancing
- Larger scale at lower latency



MRC: Modern Transport for Multi-Plane AI Fabrics

- **Adaptive Packet Spray:** Packet Multipathing for efficient Load-balancing
- **Out-Of-Order Packet Delivery**: Maximize fabric resource utilization
- **Reliable Delivery**: Efficient data transfer
- **Programmable Congestion Control**: Minimize fabric congestion
- **Advanced Telemetry**: Faster failure detection

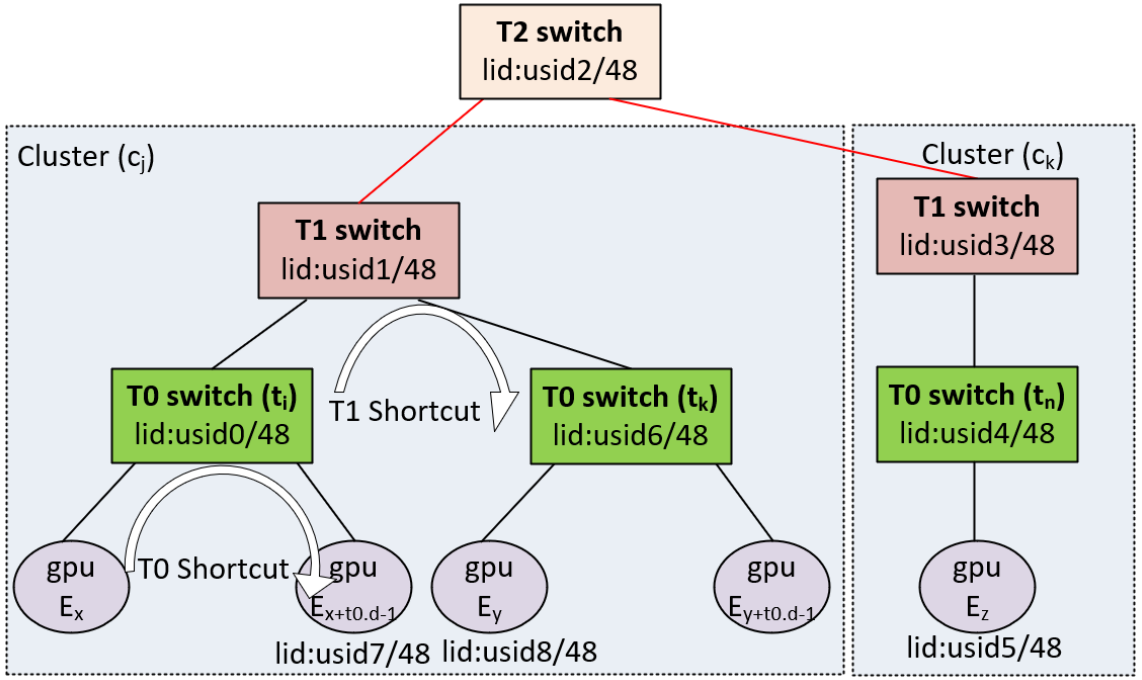
For this discussion, we will focus on Packet Multipathing and Advanced Telemetry

Packet Multipathing

- Load balancing and failure detection
  - Fine-grained path definition via structured EVs (Entropy Values) or SRv6 uSIDs
  - Active Path Failure Detection via Path Probing and management (Deny Lists, Bad paths)
- NIC distributes packets across paths and planes to a destination
  - 128 paths are tracked per QP
  - Each path is mapped to Entropy Values (EVs) carried in the packet
  - Paths may be administratively disabled (denied)
  - Path quality is communicated E2E via SACK messages
    - Simple ECN marking or Telemetry (CSIG) is supported
- Load Balancing Decision
  - A path is selected from a permutation of active paths
  - Paths are removed from permutation based on path quality
  - ECN based marked frames are skipped for a permutation

Source Routing with SRv6

- SRv6 MicroSID define the complete path through the network
  - The individual uSID values are used by the switches to determine which the next hop for the packet.
  - uSIDs are processed by each switch via a pop-and-left-shift operation.
  - The forwarding behavior is based on if uN (Micro-Node SID) or uA (Micro-Adjacency SID) mode is used.
  - SRH header is optionally used
- Example 3-tier network
  - Three hops (H1, H2 and H3) root switch



Paths

1. GPU E_x to E_z: lid:usid0:usid1:usid2:usid3:usid4:usid5
2. GPU E_x to E_{x+1}: lid:usid0:usid7:00:00:00:00 – T0 Shortcut
3. GPU E_x to E_y: lid:usid0:usid1:usid6:usid8:00:00 – T1 Shortcut

Active Path Probing

- Active Path Probing Determination
  - When a path is identified as bad, active probing is performed on the ‘bad’ paths
- Active Probing Process
  - For all paths in bad state, a probe is sent
  - A response received from a probed path puts the EV/path back to good
  - A probe is sent approximately every RTT
  - One of the bad EVs for the QP is chosen at random
  - Over time all bad EVs are probed
  - If no probe response is received that path stays bad

Advanced Telemetry

- Packet Trimming
  - AI Fabric must be TRIM enabled
  - Trimmed packet is detected using DSCP code point
  - Receiver sends NACK for each trimmed packet
  - Sender retransmits trimmed packets with priority
  - Faster detection of loss compared to timeouts - Retransmissions happen in one RTT
  - Detects congestion proactively to avoid false positives
- CSIG
  - Telemetry is gathered via per-packet fixed metadata
  - Minimum available bandwidth: Minimum available BW across all the ports in a packet path
  - Minimum available capacity: Minimum available capacity (%) across all the ports in a packet path
  - Max hop delay: Maximum delay observed across all the hops in a packet path
  - Enables scalable congestion monitoring through hardware offloaded mechanism
  - CSIG is being standardized in the UEC's Transport Working Group (TR WG)

Call to Action

- Join OCP Ethernet Projects for AI
- Open Cluster Designs for AI: https://www.opencompute.org/community/cluster-designs-for-ai
- MRC 1.0 Spec: http://www.opencompute.org/documents/ocp-mrc-1-0-pdf

Thank You!

